NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-15733/38J
4 November 1994
SUPERSEDING
MIL-F-15733/38H
12 January 1984

PERFORMANCE SPECIFICATION SHEET

FILTERS, RADIO FREQUENCY INTERFERENCE, HERMETICALLY SEALED, STYLE FL16

(J) Part numbers M15733/38-0001, -0002, -0004, -0005, are inactive for new design as of 6 October 1980. For new design use M15733/38-0006 for M15733/38-0001; use M28861/01-001TB of MIL-F-28861/1 for M15733/38-0002, -0004, and -0005.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-15733.

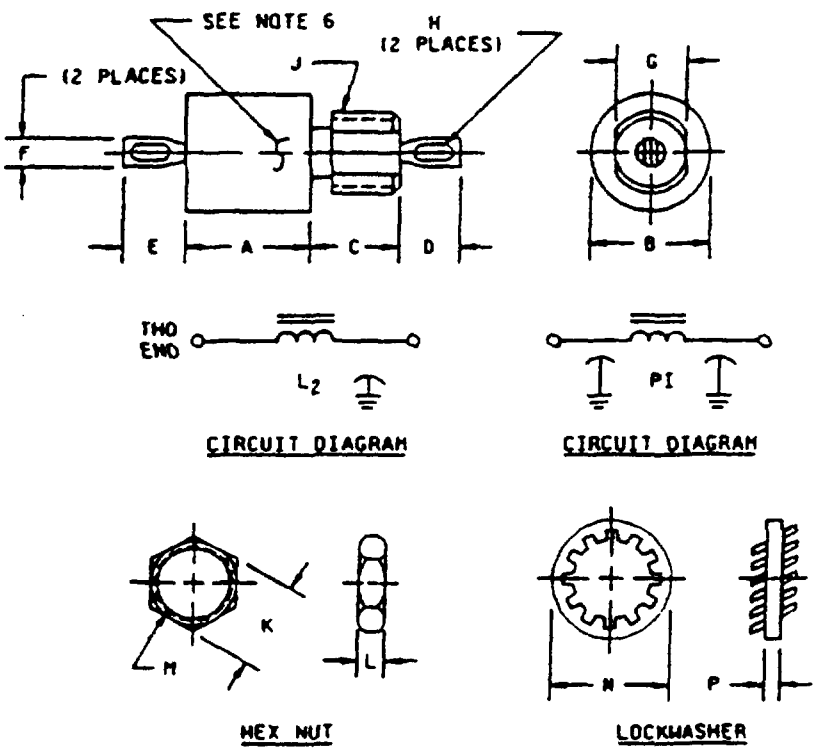


FIGURE 1. Filter and hardware dimensions and circuit diagrams.

(J) denotes changes

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Filter dimensions

Dash number	A		B		C		D		E	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
0001	.290 (7.37)	.310 (7.87)	---	.42 (10.7)	.302 (7.67)	.322 (8.18)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)
0002	.160 (4.06)	.291 (7.39)	---	.415 (10.54)	.180 (4.57)	.200 (5.08)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)
0003	.530 (13.46)	.550 (13.97)	---	.385 (9.78)	.177 (4.50)	.197 (5.00)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)
0004	.160 (4.06)	.180 (4.57)	---	.385 (9.78)	.177 (4.50)	.197 (5.00)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)
0005	.160 (4.06)	.180 (4.57)	---	.385 (9.78)	.177 (4.50)	.197 (5.00)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)
0006	.160 (4.06)	.291 (7.39)	---	.415 (10.54)	.302 (7.67)	.322 (8.18)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)
0007	.210 (5.33)	.230 (5.84)	---	.385 (9.78)	.302 (7.67)	.322 (8.18)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)
0008	.160 (4.06)	.180 (4.57)	---	.385 (9.78)	.177 (4.50)	.197 (5.00)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)
0009	.290 (7.37)	.310 (7.87)	---	.385 (9.78)	.177 (4.50)	.197 (5.00)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)
0010	.290 (7.37)	.310 (7.87)	---	.385 (9.78)	.302 (7.67)	.322 (8.18)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)

FIGURE 1. Filter and hardware dimensions and circuit diagrams - Continued.

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Filter dimensions - Continued

Dash number	F	G		H 1/		J
	+.010 -.015	Min	Max	±.010	±.010	Mounting thread
0001	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0002	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0003	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0004	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0005	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0006	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0007	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0008	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0009	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0010	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A

1/ Optional slot may be supplied .050 (1.27) ±.010 (.25) inches by .070 (1.78) ±.010 (.25) inches.

FIGURE 1. Filter and hardware dimensions and circuit diagrams - Continued.

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Hardware dimensions

Dash number	K		L		M	N		P	
	Min	Max	Min	Max	Thread	Min	Max	Min	Max
0001	.302 (7.67)	.322 (8.18)	.085 (2.16)	.095 (2.41)	.250-28 UNF-2B	.395 (10.03)	.408 (10.36)	.010 (.25)	.030 (.76)
0002	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
0003	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
0004	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
0005	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
0006	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
0007	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
0008	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
0009	.302 (7.67)	.322 (8.18)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
0010	.302 (7.67)	.322 (8.18)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagram is for information only.
4. All filters shall be supplied with mounting hardware.
5. Terminal thickness shall be .021 (0.53 mm)/.011 (0.28 mm).
6. Terminal identification. The case shall be marked at the threaded end of the filter, with the symbol "L" for dash numbers 0001, 0002, 0004, 0005, 0006, 0008, 0009, and 0010.

FIGURE 1. Filter and hardware dimensions and circuit diagrams - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 5 grams maximum.

Case: Metal.

Seal: Glass-to-metal.

Terminals: Solderable (see figure 1).

Operating temperature range:

Dash 0001 through 0006, 0008, 0009, and 0010: -55°C to +125°C.

Dash 0007: -55°C to +85°C.

Rated voltage: See table I.

Rated current: See table I.

Seal: In accordance with MIL-F-15733.

Capacitance to ground: Not specified.

Temperature rise: +25°C maximum.

Insulation resistance: In accordance with MIL-F-15733 and the following:

Test temperature: +25°C.

Test potential: Rated dc voltage or 100 volts dc whichever is less.

Insulation resistance shall be not less than:

Dash numbers	Megohms
0001	500
0002 through 0007, 0009, and 0010	100
0008	1,000

Voltage drop: See table I.

Insertion loss:

At +25°C: Shall be as specified in table I.

At -55°C and at the maximum operating temperature: Shall be as specified in table I except as follows:

Up to 10 MHz: A degradation of 4 dB from the value specified in table I shall be allowed.

Terminal strength: In accordance with MIL-F-15733 and the following:

Test condition: A (pull); applied force shall be 5 pounds.

Flashpoint of impregnant or potting compound: Not applicable.

Salt spray (corrosion): In accordance with MIL-F-15733 and the following:

Test condition: A of method 101, MIL-STD-202.

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Thermal shock and immersion: In accordance with MIL-F-15733 and the following:

Measurements after final cycling:

Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds. Maximum charging current: 50 mA.

Insulation resistance shall be not less than 30 percent of the initial requirements.

Insertion loss (check test) shall be measured and shall meet the initial requirements.

Shock (specified pulse): In accordance with MIL-F-15733 and the following:

Test condition: I (100 g).

After the test, insertion loss (check test) shall be measured and shall meet initial requirements.

Vibration, high frequency: In accordance with MIL-F-15733 and the following:

Test condition: D (20 g) of method 204, MIL-STD-202.

Life: In accordance with MIL-F-15733 and method 108 of MIL-STD-202:

Qualification inspection: Test condition D (1,000 hours).

Group C inspection:

Dash numbers	Test condition
0001, 0002, 0004 through 0010	B (250 hours)
0003	D (1,000 hours)

Part or Identifying Number (PIN): M15733/38- (dash number from table I).

Marking: Marking shall be as follows:

For dash numbers 0001, 0002, and 0004 through 0010, the filter shall be marked with the following minimum information:

a. Military PIN.

b. Source code.

c. Date code.

d. Terminal identification.

Full marking in accordance with MIL-F-15733 shall be marked on the unit package. For dash number 0003, the filter shall be marked with full marking in accordance with MIL-F-15733.

(J) Extension of qualification:

Initial qualification to MIL-F-15733/38 may be granted based on qualification to MIL-F-28861/1 or MIL-F-15733/49 as indicated in table II. Extension of qualification from MIL-F-28861/1 or MIL-F-15733/49 to MIL-F-15733/38 is permissible under the following provisions:

a. The MIL-F-15733/38 part uses the same design and dielectric characteristics as the part upon which extension of qualification is based.

b. The MIL-F-28861/1 or MIL-F-15733/49 qualification data verifies that the physical and electrical characteristics of the MIL-F-15733/38 parts are satisfied.

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TABLE I. Electrical characteristics.

Dash no.	Circuit	Max current (amps)	Voltage		Voltage drop (volts dc max)	Minimum full load insertion loss (dB) in accordance with MIL-STD-220 at +25°C 1/														
			DC volts	AC 2/ volts (rms)		30 kHz	100 kHz	150 kHz	200 kHz	300 kHz	500 kHz	1 MHz	2 MHz	5 MHz	10 MHz	20 MHz	50 MHz	100 MHz	1 GHz	10 GHz
0001	L2	10	50	-	.040	-	24	-	32	-	39	44	50	57	60	55	55	55	-	-
0002	L2	15	50	-	.12	15	-	28	-	33	-	44	-	-	60	-	-	-	70	70
0003	PI	3	150	-	.15	-	-	7	-	24	-	54	-	-	80	-	-	80	80	80
0004	L2	10	50	-	.08	15	-	28	-	33	-	44	-	-	60	-	-	-	70	-
0005	L2	10	50 3/	-	.08	15	-	28	-	33	-	44	-	-	60	-	-	-	70	-
0006	L2	15	50	-	.12	15	-	28	-	33	-	44	-	-	60	-	-	-	70	70
0007	PI	15	50	-	.09	70 dB from 420 MHz to 1.0 GHz, inclusive														
0008	L2	15	200	125	.12	-	-	-	-	-	-	6	-	-	25	-	-	-	60	-
0009	L2	15	250	125	.09	4	-	15	-	21	-	31	-	-	50	-	-	-	70	80
0010	L2	15	250	125	.09	4	-	15	-	21	-	31	-	-	50	-	-	-	70	80

1/ Full-load insertion loss measurements shall be performed over the frequency range of 100 kHz to 20 MHz inclusive. Measurements below or above this frequency range shall be performed at no-load.
2/ From dc to 400 Hz.
3/ Voltage rating at +125°C: (Note - derated from 80 V dc at +85°C).

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(J) TABLE II. Extent of qualification.

Qualification to MIL-F-28861/1	Will qualify MIL-F-15733/38	Qualification to MIL-F-15733/49	Will qualify MIL-F-15733/38
-001	-0002	-0006	-0004
-001	-0004	-0006	-0005
-001	-0005		
-011	-0006		
-009	-0008		

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - ES

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